



UNITED INTERNATIONAL UNIVERSITY
Department of Computer Science and Engineering (CSE)
Lab Evaluation 3

Course: DSA II Lab
Trimester & Year: Summer 25
Marks: 20

Course Code: CSE2218
Section A
Time: 50 min

SN	Questions	Marks						
1	You are given an array A of n positive integers and a non-negative integer K. Determine whether there exists a subset of A whose elements sum equals exactly to K using dynamic programming approach. A subset of an array means taking zero or more elements from an array. Input: The first line contains two integers n (number of elements of the array) and K (the target sum). The second line contains n integers, elements of the array. Output: “True” if any subset sum equals to K, or “False” otherwise. <table><tr><th>Sample Input</th><th>Sample Output</th></tr><tr><td>5 10 2 4 5 6 8</td><td>True</td></tr><tr><td>5 3 2 4 5 6 8</td><td>False</td></tr></table> Explanation: For Sample Input 1, one valid subset is {4, 6} which sums to 10. So, the answer is True. For Sample Input 2, there is no valid subset which sums to 3. So, the answer is False. Hint: Think like the coin change problem, and save True / False in the dp array instead of number of coins. Also think when an element of the dp array will be true and update accordingly.	Sample Input	Sample Output	5 10 2 4 5 6 8	True	5 3 2 4 5 6 8	False	10
Sample Input	Sample Output							
5 10 2 4 5 6 8	True							
5 3 2 4 5 6 8	False							
2	You are climbing a staircase that has n steps. Each move you may climb 1, 2, or 3 steps. Count the number of distinct ways to reach the top (step n) from the bottom (step 0) using dynamic programming approach. Input: An integer n representing the number of steps. Output: A single integer denoting the number of distinct ways to reach step n. <table><tr><th>Sample Input</th><th>Sample Output</th></tr><tr><td>3</td><td>4</td></tr><tr><td>4</td><td>7</td></tr></table> Explanation: For Sample Input 1, the distinct ways to climb are (1+1+1), (2+1), (1+2), (3). So, the answer is 4. For Sample Input 2, the distinct ways to climb are (1+1+1+1), (2+1+1), (1+2+1), (1+1+2), (2+2), (3+1), (1+3). So, the answer is 7. Hint: Use bottom up dp. Think for each n how it can be calculated from the previous steps.	Sample Input	Sample Output	3	4	4	7	10
Sample Input	Sample Output							
3	4							
4	7							